SENYAS: An Offline Intelligent System for Trilingual Sign Language and Braille Translation Using a Kinematic Hand Model, Landmark Geometry, and Braille Grid Fitting

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ABSTRACT

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Deaf, blind and hearing Filipinos often cannot talk to one another without an interpreter, and in Albay no tool works in Bikol, offline, on an ordinary phone. This study designed, built and evaluated SENYAS, an intelligent system that reads Filipino Sign Language (FSL) fingerspelling and everyday signs through a camera, speaks the words when the user switches reading aloud on, signs typed or spoken English, Filipino and Bikol back with an avatar, reads braille from a photo or the live camera, and teaches signing. Instead of a large video dataset, it uses the kinematics of a hand modelled like a robot hand to generate its examples, palm-frame coordinates that do not change with position, size or rotation, a nearest-neighbour rule, hysteresis counting and winding numbers for movement, dynamic time warping for signs the user teaches, and a spacing-error fit that snaps braille dots onto their grid. In repeatable computational experiments, static letters were named with 99.0% accuracy at expected camera noise; 570 of 570 moving and two-handed sign trials were detected with no false alarms on still hands; 89.2% of fingerspelled words were read exactly at the default setting; signs taught from three recordings were found in 120 of 120 trials with no false accepts; and 81.7% of 300 braille photos were read exactly (12.6% character error rate). Each frame needed 71 ms for hand tracking and 0.09 ms for recognition, and the interface fitted all 11 screen sizes tested. SENYAS shows that geometry and kinematics can give an offline, trilingual communication aid without big data. A field validation with Deaf and blind learners in Albay is recommended.

Keywords: intelligent machines, computer vision, hand kinematics, Filipino Sign Language, braille recognition, assistive technology

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CHAPTER 1. INTRODUCTION

1.1 Background of the Study

Republic Act No. 11106 (2018) declared Filipino Sign Language (FSL) the national sign language of the Filipino Deaf and required its use in schools, broadcasts, courts and public health services. Two years earlier, the World Health Organization had begun warning that hearing loss was growing; its World Report on Hearing estimates that more than 430 million people need care for disabling hearing loss (World Health Organization, 2021), and its World Report on Vision counts at least 2.2 billion people with a vision impairment (World Health Organization, 2019). The Magna Carta for Disabled Persons (Republic Act No. 7277, 1992) and the Convention on the Rights of Persons with Disabilities (United Nations, 2006) promise equal access to information. Yet the law and the numbers do not open a conversation by themselves. A Deaf student who needs directions at a terminal, a blind grandmother holding a printed notice, or a hearing nurse explaining a medicine still depends on an interpreter or a reader who is rarely there.

Albay adds two barriers of its own. First, most translation tools work in English, some in Filipino, and almost none in Bikol, the language most families in the province speak at home. Second, the moments when communication matters most, such as an evacuation before Mayon erupts, a typhoon or an earthquake, or a consultation in a far barangay health station, are exactly when mobile data fail. A tool that needs a server stops working at the moment it is needed.

Phones can now track the 21 joints of each hand in real time without the internet (Zhang et al., 2020). What remains is machine intelligence: turning those points into letters and words reliably, recognising signs that move, reading braille dots from a picture taken at an angle, and doing all of this in three languages on an inexpensive device. This is a problem for intelligent machines in the full sense: the system must perceive (camera), model (a kinematic hand), decide (classification under noise), act (speak, sign with an avatar) and adapt (learn new signs from the user). This study builds and tests such a system, called SENYAS, from the Filipino word for sign or gesture.

Current research on sign language recognition is dominated by deep neural networks trained on thousands of annotated videos (Koller, 2020; Rastgoo et al., 2021). Such networks reach high accuracy on the languages and signers they were trained on, but FSL, and FSL used by Bikol

speakers, has no public dataset of that size, and the networks are heavy to run offline on low-cost phones. Optical braille recognition has a similar history: published systems rely on flatbed scanners and controlled lighting (Antonacopoulos & Bridson, 2004; Isayed & Tahboub, 2015), not on a phone held over a page. A different route is to describe the hand and the braille cell with geometry. A hand is a chain of rigid links joined by revolute joints, exactly like a robot hand (Craig, 2005); a braille cell is a two-by-three grid with fixed spacing (International Council on English Braille, 2013). If these structures are written as equations, labelled examples can be generated instead of collected, features can be made independent of how the hand or page is held, and every decision can be explained and tested. SENYAS follows this route.

1.2 Statement of the Problem

The study aimed to design, develop and evaluate SENYAS, an offline intelligent system that translates between sign language, braille, text and speech in English, Filipino and Bikol, with a learning module. Specifically, it sought to answer the following:

1. How accurately does a nearest-neighbour recogniser that uses palm-frame features, with examples generated by a kinematic hand model, classify the 24 static letters of the FSL manual alphabet as pose noise increases, and which design components account for its accuracy?
2. How reliably do the motion rules and two-hand rules detect 13 moving and 6 two-handed signs, and how often do still letters trigger a false moving sign?
3. How accurately are whole fingerspelled words read, and how does the hold-time setting interact with the signer's speed?
4. How reliably are signs that the user teaches from three recordings recognised, and how often are other signs falsely accepted as taught signs?
5. How accurately does the grid-fitting braille reader decode photos of Grade 1 braille that differ in dot type, skew, blur and dot size?
6. How long does each component take per camera frame or photo, and is this fast enough for live use?

7. Does the interface adapt to phones, tablets, computers and televisions without layout faults, with readable contrast and adequate touch targets, and how completely does the trilingual lexicon carry the phrasebook into signs?

1.3 Hypotheses

The following null hypotheses were tested at the 0.05 level of significance.

Table 1. Null hypotheses and their tests

Code	Null hypothesis	Test
$H_0 1$	Static-letter accuracy at expected noise is at most 90%.	One-sided exact binomial test
$H_0 2$	The detection rate of moving and two-handed signs is at most 90%.	One-sided exact binomial test
$H_0 3$	Fingerspelled-word accuracy at the default hold time is at most 80%.	One-sided exact binomial test
$H_0 4$	The braille reader's character error rate is at least 20%.	Comparison with the bound; Wilson interval
$H_0 5$	Processing time per frame exceeds 100 ms (fewer than 10 frames per second).	Median latency
$H_0 6$	The interface shows sideways scrolling or a missing navigation bar on at least one tested screen size.	Automated layout test

1.4 Significance of the Study

Deaf and hard-of-hearing learners. SENYAS turns their signing into spoken Filipino, English or Bikol and lets them follow hearing speakers through an avatar, at no cost and without data.

Blind and low-vision persons. Braille labels, notices and school materials can be read aloud from a photo, and typed text is shown in braille for learners.

Teachers, SPED centres and families. Lessons, quizzes, flashcards, games and camera practice support learning FSL in the home language, in line with Universal Design for Learning (CAST, 2018).

Health workers, police and disaster responders. An emergency phrasebook (help, doctor, hospital, fire, earthquake, flood, typhoon) works with no signal during disasters in Albay.

Future researchers. The study offers an explainable method that does not need a large dataset and open benchmarks that can be repeated and extended.

The study supports Sustainable Development Goals 4 (inclusive and quality education), 10 (reduced inequalities) and 11.5 (reducing the effects of disasters) (United Nations, 2015), and the aims of RA 11106 and RA 7277.

1.5 Scope and Delimitation

The study covers the one-handed manual alphabet A–Z with Ñ, the numbers 0–10, and 21 everyday signs read directly by the camera, in forms shared by FSL and American Sign Language; 93 signed words and 32 phrases in English, Filipino and Central Bikol; signs taught by the user; Grade 1 (uncontracted) braille with capitals, numbers, common punctuation, Ñ and five whole-word contractions; and the web, installable and Android versions of the app. The study does not cover full FSL sentences with grammar and facial expressions, Grade 2 braille, or testing with Deaf or blind participants, which is recommended for the next phase. All performance figures come from computational experiments on generated data, which are described fully so that they can be repeated. The rendered-hand test of Section 4.8 is a lower bound on the full camera path, not a field accuracy.

1.6 Definition of Terms

Table 2. Definition of terms

Term	Conceptual meaning	Operational meaning in this study
Intelligent system	A machine that perceives, decides and acts toward a goal.	SENYAS: camera and microphone input, recognition, speech and avatar output.
Landmark	A point that marks a body joint.	One of the 21 hand points reported by MediaPipe for each hand.
Kinematic hand model	Equations that place the links of a hand from its joint angles.	The SENYAS model that builds 37 handshapes for templates and the avatar.
Palm frame	A coordinate system attached to an object.	Axes built from the wrist and three knuckles of a detected hand.
Template	A stored example of a class.	A 41-number feature vector of a known handshape.
Fingerspelling	Spelling words with handshapes of the alphabet.	Letters held for the hold time are added to the current word.
Hold time	How long a handshape must stay before it counts.	300, 400 (default) or 600 ms in the app settings.
Braille cell	Six dot positions in two columns and three rows.	A six-bit pattern decoded into a letter, digit or sign.
Character error rate	The share of characters read wrongly.	Levenshtein distance divided by the length of the true text.
Read aloud	Speaking recognised text.	The switch that makes SENYAS speak each recognised word; the user turns it on or off.

CHAPTER 2. REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Sign Language and Filipino Sign Language

Sign languages are full natural languages with their own grammar, not codes for spoken languages. Stokoe (1960) showed that each sign can be described by a small set of parameters: the handshape, the location where it is made, and the movement. This description is the basis of SENYAS: the handshape is read from the joint positions, the location from where the hand is in the frame and relative to the other hand, and the movement from how these change over time. RA 11106 (2018) makes FSL the medium of instruction for Deaf learners and the language of interpreting in government. FSL shares its one-handed manual alphabet with ASL, so a fingerspelling recogniser serves users of both, while many FSL words have local forms that differ by region.

2.2 Communication Access for Deaf and Blind Filipinos

The World Health Organization (2019, 2021) reports that hearing and vision impairments are common and rising, and that most affected people live in low- and middle-income countries where services are scarce. Philippine law guarantees access: RA 7277 (1992) requires auxiliary aids and accessible communication, and the Data Privacy Act (RA 10173, 2012) protects the personal information that any assistive tool might collect. Interpreters and braille transcribers remain few, especially outside Metro Manila, so most daily conversations between Deaf, blind and hearing people happen without support.

2.3 Vision-Based Sign Language Recognition

Early gesture recognisers used data gloves or coloured markers; camera-based methods later replaced them (Mitra & Acharya, 2007). Surveys of the field show that recent systems use convolutional and recurrent networks or transformers trained on large datasets, and that results drop sharply on new signers or new sign languages (Koller, 2020; Rastgoo et al., 2021). For FSL, Cabalfin et al. (2012) recognised signs from video using manifold projection learning, showing that FSL recognition by computer is possible but also that data are limited. Landmark-based methods reduce the input to joint positions, which removes background and skin colour from the problem and makes small models possible.

2.4 Hand Tracking and Kinematics

MediaPipe is a framework for building perception pipelines that run on devices (Lugaresi et al., 2019). Its hand solution first detects a palm and then regresses 21 three-dimensional landmarks in real time on mobile hardware (Zhang et al., 2020). It reports both image coordinates and world coordinates in metres around the hand. In robotics, the position of a manipulator's links follows from its joint angles through forward kinematics, and the joint angles that place its end at a target follow from inverse kinematics (Craig, 2005). FABRIK solves inverse kinematics by alternately pulling a chain toward its target and back to its base, converging quickly without matrix inversion (Aristidou & Lasenby, 2011). A human hand can be modelled with the same tools: four fingers with three revolute joints each and a thumb with a more mobile base.

2.5 Pattern Recognition over Time

The nearest-neighbour rule assigns an input to the class of its closest stored example; Cover and Hart (1967) proved that, as examples grow, its error is at most twice the lowest possible (Bayes) error. Dynamic time warping aligns two sequences that unfold at different speeds, and a Sakoe–Chiba band limits how far the alignment may stretch, which both speeds it up and prevents unrealistic matches (Sakoe & Chiba, 1978). Hysteresis, first used in the Schmitt (1938) trigger, makes a detector change state only after a signal moves past a margin, so small jitter does not cause false switching; SENYAS uses the same idea to count the back-and-forth of a moving hand.

2.6 Braille and Optical Braille Recognition

In Unified English Braille, each letter is a pattern of up to six raised dots in a cell; a number sign turns the letters a–j into digits and a capital sign marks capitals (International Council on English Braille, 2013). Filipino uses the same letters with Ñ. Optical braille recognition must find dots that may be printed or embossed, lit from one side, and slightly skewed. Antonacopoulos and Bridson (2004) recognised braille from scanned pages by finding dot shadows and fitting a grid, and Isayed and Tahboub (2015) found that grid estimation and skew are the hardest steps. Summed-area tables give the mean of any rectangle in constant time (Crow, 1984; Viola & Jones, 2001), which makes local contrast fast enough for live video.

2.7 Accessible and Adaptive Software

The Web Content Accessibility Guidelines 2.1 set measurable criteria, including a contrast of at least 4.5 to 1 for normal text and content that reflows without sideways scrolling (World Wide Web Consortium, 2018). Responsive design adapts one layout to many screens (Marcotte, 2010). ISO/IEC 25010 (2011) defines software quality in eight characteristics, from functional suitability and performance efficiency to usability, security and portability, which this study uses to organise its evaluation. Universal Design for Learning recommends many means of representation and expression (CAST, 2018), which the learning module follows by showing every sign as an animation, three written words, speech and, for letters, braille.

2.8 Related Studies

Table 3. Related studies and their relation to the present study

Study	What was done	Relation to the present study
Zhang et al. (2020)	Real-time 21-point hand tracking on phones.	Supplies the landmarks SENYAS reads.
Cabalfin et al. (2012)	FSL recognition from video with manifold learning.	Confirms FSL recognition is possible; SENYAS adds offline use, Bikol and braille.
Koller (2020); Rastgoo et al. (2021)	Surveys: most methods are deep and data-hungry.	Motivates a geometric method that needs no large dataset.
Mitra & Acharya (2007)	Survey of gesture recognition methods.	Supports separating handshape from movement.
Aristidou & Lasenby (2011)	FABRIK inverse kinematics.	Places the thumb of the hand model and the avatar.
Antonacopoulos & Bridson (2004)	Braille recognition from scanned pages.	SENYAS reads phone photos and live video with skew and blur.
Isayed & Tahboub (2015)	Review of optical braille recognition.	Identifies grid fitting and skew as key steps, addressed here.

2.9 Synthesis and Research Gap

Hand tracking is good enough to run on phones, but no study has turned landmarks into FSL letters and words offline, with Bikol output and speech, and joined this with braille reading and an avatar that signs back. Deep methods need data that FSL does not have; braille readers assume scanners. The gap this study fills is an intelligent, explainable, offline system for the three languages of Albay whose behaviour can be stated, measured and improved.

2.10 Theoretical Framework

Table 4. Research gaps and the theories used to address them

Gap	Theory or method that addresses it
G1 No large FSL dataset	Robot-hand kinematics (forward kinematics and FABRIK) generates

	labelled handshapes (Craig, 2005; Aristidou & Lasenby, 2011).
G2 Hands differ in position, size and angle	Coordinates in a frame attached to the palm are unchanged by translation, scaling and rotation.
G3 Noisy landmarks and similar shapes	Nearest-neighbour classification with a confidence margin (Cover & Hart, 1967).
G4 Signs move at different speeds	Hysteresis counting (Schmitt, 1938), winding numbers, and dynamic time warping (Sakoe & Chiba, 1978).
G5 Braille photos are skewed and noisy	Summed-area tables (Crow, 1984), robust spacing and skew estimates, and grid fitting by error minimisation.
G6 Three languages, no signal, many screens	A trilingual phrase index with longest-match lookup; responsive, accessible design (W3C, 2018).

2.11 Conceptual Framework

Figure 1 shows the input–process–output paradigm. The inputs are camera frames of hands, photos or frames of braille, and typed or spoken text. The processes turn them into landmarks and features, decisions about signs, dots and cells, and lookups in the trilingual index. The outputs are text in the chosen language with optional speech, and an avatar that signs, with the text also shown in braille. Benchmark results feed back into the design.

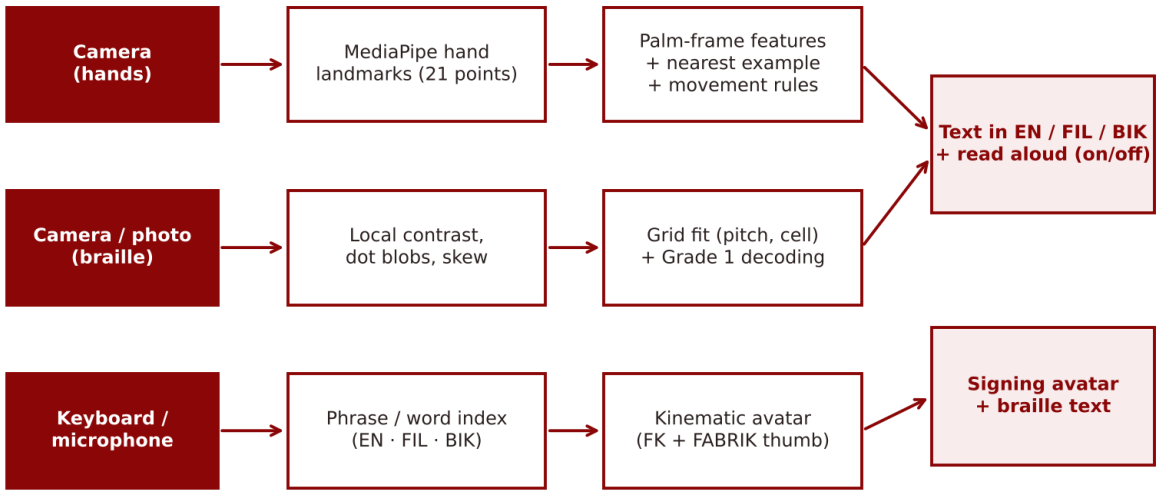


Figure 1. Conceptual paradigm of the study

CHAPTER 3. METHODOLOGY

3.1 Research Design

The study used a developmental engineering-design approach (Hevner et al., 2004). The intelligent system was designed from kinematic and geometric models, built as working software

for phones and computers, and evaluated quantitatively through nine repeatable computational experiments that simulate its inputs. The independent variables were pose noise, the sign, the hold time and signer speed, the braille photo condition and the screen size; the dependent variables were accuracy, detection and false-alarm rates, word accuracy, character error rate, processing time and layout faults.

3.2 Materials

Table 5. Materials used in development and testing

Item	Specification	Use
Computer	x86-64 CPU, Linux; no GPU used in tests	Development and all benchmarks
Chromium (headless) with Playwright	Desktop browser engine and test driver	Runs the app, MediaPipe, braille and layout tests
Node.js 22	JavaScript runtime	Recogniser benchmarks
MediaPipe Tasks Vision 0.10.14	Hand Landmarker, float16 model (Apache-2.0)	On-device hand landmarks
HTML, CSS and JavaScript	No framework; one 190 KB page	The app and the website
Kotlin and Android WebView	Android 7.0 or later	Android app with phone speech and camera
Python 3, Matplotlib, python-docx	Statistics, figures, documents	Analysis and reporting

3.3 System Architecture

SENYAS has four engines that share one page: the sign engine (camera → MediaPipe → features → decisions), the avatar engine (text → index → signs → kinematic animation), the braille engine (photo or frame → dots → grid → text), and the speech engine (text → device voice, and microphone → text). All engines run on the device. A consent sheet decides what is saved; video, photos and audio are never stored or sent. The app adapts its layout with CSS media queries: a bottom tab bar on phones, a side rail on tablets, both translation directions side by side on wide screens, a camera-beside-text layout on phones held sideways, and larger type on televisions.

3.4 Development Procedure

3.4.1 The kinematic hand model

Each finger i has a knuckle at m_i on the palm, a spread angle a_i and three links of lengths ℓ_1, ℓ_2, ℓ_3. With joint flexions θ_1, θ_2, θ_3, link j points in the direction

$$d_j = \cos(\theta_1 + \ldots + \theta_j)\, u + \sin(\theta_1 + \ldots + \theta_j)\, z, \quad u = (\sin a_i, \cos a_i, 0) \tag{1}$$

where z is the palm normal, so joint j lies at $m_i + \ell_1 d_1 + \dots + \ell_j d_j$. This is forward kinematics for a planar three-link chain turned by the spread angle. The thumb has a more mobile base, so it is placed by FABRIK: its base is fixed, its tip is sent to a target (for example, the side of the index finger for A, or the tip of the middle finger for D), and the chain is pulled alternately to the target and the base for 14 iterations. A handshape is therefore a vector of 12 finger angles, 4 spreads and one thumb target; 37 shapes were defined this way. An orientation is given by the finger direction f and palm direction n, made orthonormal and assembled into the rotation [f × n, f, n]. The same model animates the avatar, blending angles, thumb targets and orientations between keyframes with a smoothstep curve, so the shapes it signs are the shapes it recognises.

3.4.2 Palm-frame features

For the landmarks $p_0 \dots p_{20}$ of a detected hand (wrist p_0; index, middle and little-finger knuckles p_5, p_9, p_{17}), the palm frame is

$$s = \|p_9 - p_0\|, \quad Y = (p_9 - p_0)/s, \quad Z = (p_5 - p_{17}) \times Y \,/\, \|(p_5 - p_{17}) \times Y\|, \quad X = Y \times Z \tag{2}$$

and each fingertip and middle joint q is written as $c(q) = ((q - p_0)\cdot X, (q - p_0)\cdot Y, (q - p_0)\cdot Z)/s$. These 30 values are joined by four thumb-to-fingertip distances, three distances between neighbouring fingertips and four orientation terms, giving a weighted 41-number vector.

Proposition 1 (invariance). If every landmark is moved by a similarity transform $p \mapsto \lambda R p + t$ with $\lambda > 0$ and R a rotation, the 30 palm-frame values and the 7 distance ratios do not change.

Proof. The palm length becomes λs. The differences $p_9 - p_0$ and $p_5 - p_{17}$ become $\lambda R(p_9 - p_0)$ and $\lambda R(p_5 - p_{17})$; since a rotation preserves lengths and cross products, the new axes are RX, RY and RZ. For any joint, $(\lambda R(q - p_0))\cdot(RY)/(\lambda s) = (q - p_0)\cdot Y/s$ because $R^T R = I$, and likewise for X and Z. Distance ratios are unchanged because all lengths scale by λ. ∎

A reflection has determinant −1 and reverses the cross product, so the mirror image of a right hand has its Z axis flipped; this is why a left hand is mirrored into a right hand before comparison. Because a camera's left/right label can be wrong, SENYAS compares both the hand and its mirror image and keeps the closer match. The four orientation terms are deliberately not invariant: they tell apart letters that share a handshape but point differently, such as G and Q, K and P, and H and U.

3.4.3 Templates, classification and confidence

Each built-in template is a hand-model shape turned by random rotations of up to about ±17° and loosened by random joint changes, giving 572 templates. A new hand is assigned the label of its nearest template by Euclidean distance; examples that the user records are weighted by 0.8 so that they are preferred. With d_1 and d_2 the distances to the best and second-best labels, the confidence is

$$c = \mathrm{clip}((1.55 - d_1)/1.0,\ 0,\ 1) \times \min(1,\ 0.35 + 2.2\,(d_2 - d_1)) \tag{3}$$

Classifying one hand costs about N × D = 572 × 41 ≈ 23,000 multiply-adds, which is negligible on a phone.

3.4.4 Decisions over time

A letter is written only when the same label has been held for the hold time (400 ms by default) while the wrist moves less than 0.6 palm-lengths over 0.4 s, so letters are not written during transitions. A double letter, as in HELLO, is signed with a short sideways slide of the same handshape: when the wrist has moved more than 0.4 palm-lengths since the last letter and then rests, the letter is written again. When no hand is seen for 0.9 s, the spelled word is closed and looked up in the dictionary, so that spelling T-U-B-I-G gives water, and the word is spoken if reading aloud is on.

3.4.5 Moving and two-handed signs

The last 1.5 s of wrist and fingertip positions are kept in palm-length units. Back-and-forth movement is counted with hysteresis: a reversal is counted only after the signal has turned back by more than an amplitude A, so jitter smaller than A is ignored. Hello is an open hand whose fingertips reverse at least twice; yes is a fist whose knuckles reverse vertically; where is a wagging index finger; bathroom is a shaking T; water and eat are taps of W and flat-O hands. Please and sorry are circles of a flat hand and a fist, detected by the winding number

$$w = (1/2\pi)\,\Sigma_t\ \mathrm{wrap}(\varphi_t - \varphi_{t-1}),\quad \varphi_t = \mathrm{atan2}(y_t - \bar{y},\ x_t - \bar{x}) \tag{4}$$

which must exceed 0.8 turns, with a mean radius above 0.28 palm-lengths and a roundness (short side over long side of the path's bounding box) above 0.45. Thank you is a flat hand moving down from chin height. J, Z and Ñ are the I, index and N handshapes held throughout a hook, a zig-zag and a shake. Two-handed signs are named from the pair of handshapes and the distance

between chosen points: two fists close together for love, a thumbs-up resting above a flat palm for help, touching flat-O fingertips for more, touching flat-hand fingertips for house, linked hooked index fingers for friend, and tapping H hands for name.

3.4.6 Signs taught by the user

A user can teach any sign, such as a local FSL word or a name sign, by recording it three times for 1.5 s. If the hand moves less than 1.1 palm-lengths, the three mean feature vectors become templates. Otherwise each take is resampled to 16 frames of 44 numbers (the features plus the wrist displacement and whether a second hand is present) and stored as a sequence. A new movement is compared with each take by dynamic time warping with a band of four frames; its distance is divided by a tolerance τ equal to 1.6 times the average distance between the user's own three takes (bounded to 0.75–1.6), and the sign is accepted when the ratio is below 1. Each comparison costs about 16 × 9 × 44 operations.

3.4.7 The braille reader

A photo or video frame is reduced to at most 1,100 pixels wide and turned to grey. A summed-area table gives the local mean over a window of about one-fortieth of the image in constant time per pixel; pixels darker than the local mean by 1.8 standard deviations (printed dots) or lighter (embossed dots catching the light) form candidate blobs found by flood fill. Blobs that are smaller than 8 pixels, elongated or far from the typical dot size are discarded, and the polarity whose blobs are most uniform and best supported by neighbours at dot spacing is kept. The dot spacing δ is the median of the nearest-neighbour distances that are within 1.2 times their lower quartile, which ignores the longer diagonal and between-cell distances, and the skew is the median nearest-neighbour angle folded into (−45°, 45°]. After the points are turned upright, rows are clustered with a gap of δ/2 and grouped into lines of three. In each line the cell grid is found by minimising

$$E(P, x_0) = \Sigma_c \min(|u_c - kP|, |u_c - \delta - k'P|), \quad u_c = x_c - x_0, \quad 2.1\delta \le P \le 2.9\delta \tag{5}$$

over the cell pitch P and offset x_0, where x_c are the column centres and k, k′ the nearest integers. Each dot is then given a cell, column and row, forming a six-bit number that a small state machine decodes: the number sign switches the letters a–j to digits until a space, the capital sign raises the next letter, and standalone cells for and, for, of, the and with are read as words. The whole reader is linear in the number of pixels apart from the small grid search.

3.4.8 Translation, speech and the avatar

All 93 words, the numbers and 32 phrases are stored with their English, Filipino and Bikol forms in a single index. Text is normalised (lower case, accents removed) and matched greedily, longest phrase first up to seven words; matched phrases expand into their sign order, single words map to their signs, and anything else is fingerspelled. The avatar is a drawn signer whose arms are placed by two-link inverse kinematics from the shoulder to the wrist, with each hand drawn from the kinematic model with depth ordering, simple facial expressions for questions and feelings, and adjustable speed. Speech uses the device voice: on the web, the Speech Synthesis API; on Android, the phone's text-to-speech engine through a JavaScript bridge. Bikol text is read with a Filipino voice, the closest one available.

3.4.9 The app, website and privacy design

The app is a single installable web page (190 KB, plus the 17 MB hand-tracking engine kept for offline use) that works on Android, iPhone, iPad, computers and televisions; an Android app wraps the same page with access to the camera, microphone and phone voice, and is built automatically for download from the website. The website explains the features, shows a live avatar demonstration, offers downloads for each device and includes a cookie banner and privacy policy. In the app, a consent sheet explains that video, photos and audio never leave the device and lets the user choose whether settings, progress and taught signs are saved; with essential storage only, nothing else is kept.

Algorithm 1. Reading one camera frame

```
for each detected hand h:
    f1 ← features(h as labelled), f2 ← features(mirror of h)
    d[label] ← min over templates of min(‖f1 − t‖, 1.04‖f2 − t‖)
choose the dominant hand; append its points to the 1.5 s buffer
if a taught moving sign matches (DTW ratio < 1): write it; stop
if a moving-sign rule fires: write the sign; clear the buffer; stop
if two hands form a two-handed sign for the hold time: write it; stop
best ← nearest allowed label; if confidence > 0.32 and steady for the hold time:
    write the letter or word (speak it if Read aloud is on)
```

3.5 Data Gathering Procedure

Table 6. The nine experiments

No.	Experiment	Data generated	Measure
E1	Static letters	24 letters × 40 hands × 4 noise levels (960 per level); one in four left-handed	Accuracy
E2	Ablation	24 letters × 30 hands at 1.5× noise, 15% wrong hand	Accuracy

		labels, five designs	
E3	Moving and two-handed signs	19 signs × 30 streams at 30 frames per second with random size and speed; 144 still streams	Detection, false alarms
E4	Fingerspelled words	40 words × 3 hold times × 3 signer speeds, with transitions and double letters	Word accuracy, CER
E5	Taught signs	4 new movements, 3 recorded takes each, 30 new trials each, 104 look-alike trials	Detection, false accepts
E6	Braille photos	300 photos with random dot type, skew ±6°, spacing 12–20 px, blur 0–1.6 px, uneven light, noise; ablation of two steps	Exact rate, CER
E7	Full camera path	288 rendered hands (24 letters × 3 skin tones × 2 hands × 2 tilts) given to MediaPipe	Detection, accuracy
E8	Speed	400 recogniser frames, 288 MediaPipe images, 300 braille photos	Median time
E9	Screens and access	11 screen sizes × 5 screens; 7 colour pairs; phrasebook in 3 languages	Layout faults, contrast, coverage

In E1–E5, hands were generated by the kinematic model with random joint changes (standard deviation 7° times the noise multiplier), thumb-target changes, rotations of 0.2–0.3 radians times the multiplier, and landmark noise of 2.5–3.5 mm times the multiplier, then converted to MediaPipe's coordinate conventions and passed through exactly the same code the app runs. Random numbers came from a fixed-seed generator, so every run can be repeated.

3.6 Treatment of Data

Proportions are reported with 95% Wilson (1927) intervals, which remain accurate near 0% and 100%. $H_0$1–$H_0$3 were tested with one-sided exact binomial tests, $p = P(X \geq k \mid n, p_0)$. The character error rate is the total Levenshtein (1966) distance divided by the total length of the true text, with each photo's error capped at its length. Times are medians. All computations are in the open scripts bench_recognizer.js, bench_browser.js, bench_ui.js and analyze.py.

3.7 Ethical Considerations and Safety

No human participants took part in this phase, so no personal data or consent forms were needed. The app processes camera images, photos and audio on the device and never uploads them; it asks before saving anything, lets users export or delete their data, and follows the Data Privacy Act of 2012. The app tells users that its signs are forms shared by FSL and ASL, that local FSL signs vary, and that it is not a certified interpreter; for medical, legal and emergency matters it advises asking for a qualified FSL interpreter. All sources are cited, and the open-source hand-tracking engine is used under its Apache 2.0 licence. Any future test with Deaf or blind persons

will undergo ethics review, obtain parental consent for minors, and be carried out with the Deaf community.

CHAPTER 4. RESULTS AND DISCUSSION

4.1 The Developed System

The study produced a working system with five parts: (1) live sign-to-words translation with a Read aloud switch, letter, number and word modes, and a conversation history; (2) words-and-voice-to-sign with the avatar, speed control, quick phrases and a braille line; (3) a braille reader for photos and live video with a braille chart; (4) a learning module of 11 units with lessons, quizzes, camera practice, flashcards, two games, levels, streaks and badges; and (5) a dictionary and phrasebook in three languages. Settings include the app language, output language, voice speed, signing hand, camera, hold time, avatar skin tone, theme, text size, high contrast, reduced motion, taught signs and data controls. Figure 2 shows the main screens.

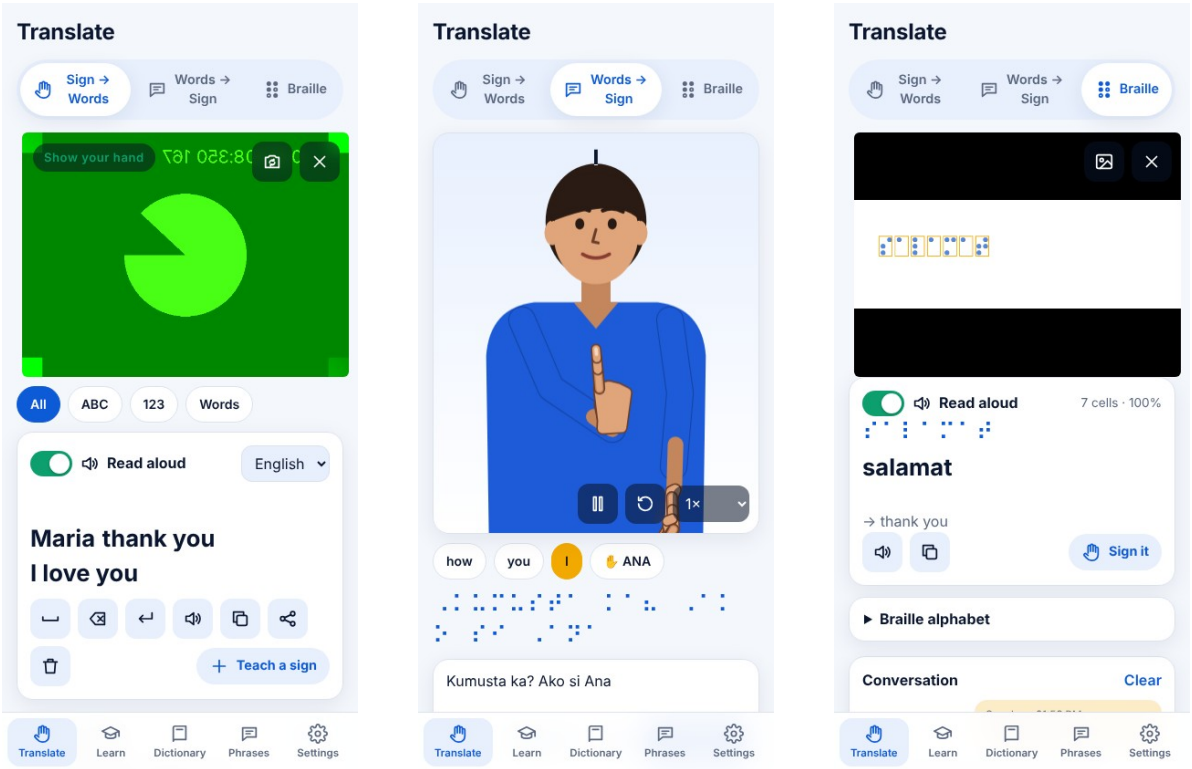


Figure 2. Live sign translation with Read aloud, the signing avatar, and the braille reader on a phone

4.2 Static Letters (E1)

Table 7. Accuracy on the 24 static letters

Noise multiplier	Correct / trials	Accuracy	95% Wilson CI

0.5×	959 / 960	99.9%	99.4% – 100.0%
1×	950 / 960	99.0%	98.1% – 99.4%
1.5×	903 / 960	94.1%	92.4% – 95.4%
2×	853 / 960	88.9%	86.7% – 90.7%

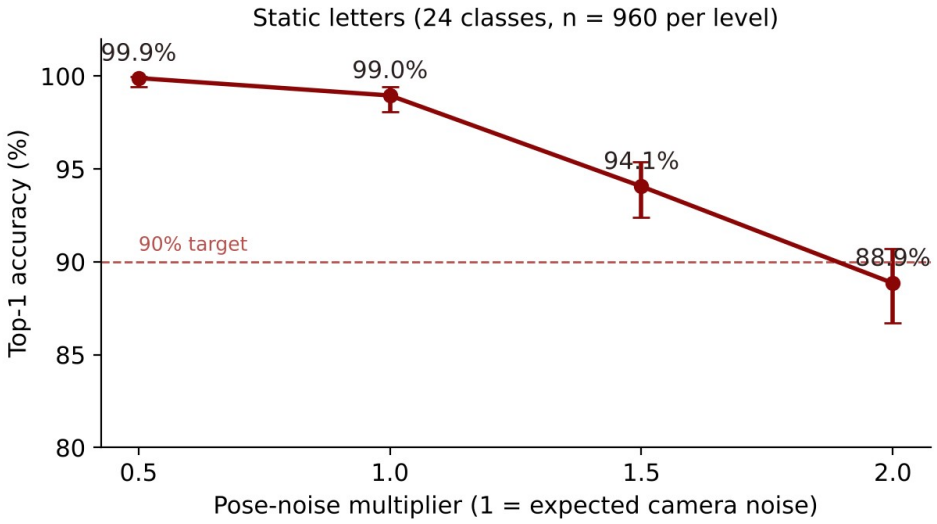


Figure 3. Static-letter accuracy as pose noise increases

At the expected camera noise the recogniser named 950 of 960 hands correctly (99.0%); the exact binomial test rejected $H_0 1$ ($p < 0.001$). Accuracy declined gradually, to 88.9% at twice the expected noise, instead of collapsing. Figure 4 shows where the errors went at 1.5 times the noise. The letters with the lowest accuracy were U (63%), N (82%), R (82%), T (88%), and the most frequent confusions were U→R (34), R→U (21), V→U (15), N→M (12), G→Q (10). These pairs are close for human readers as well: U and R differ only in whether the fingers cross, G and Q only in pointing down, and S, T, M and N only in where the thumb rests. This is why the app lets each user record personal examples of hard letters, which the classifier prefers.

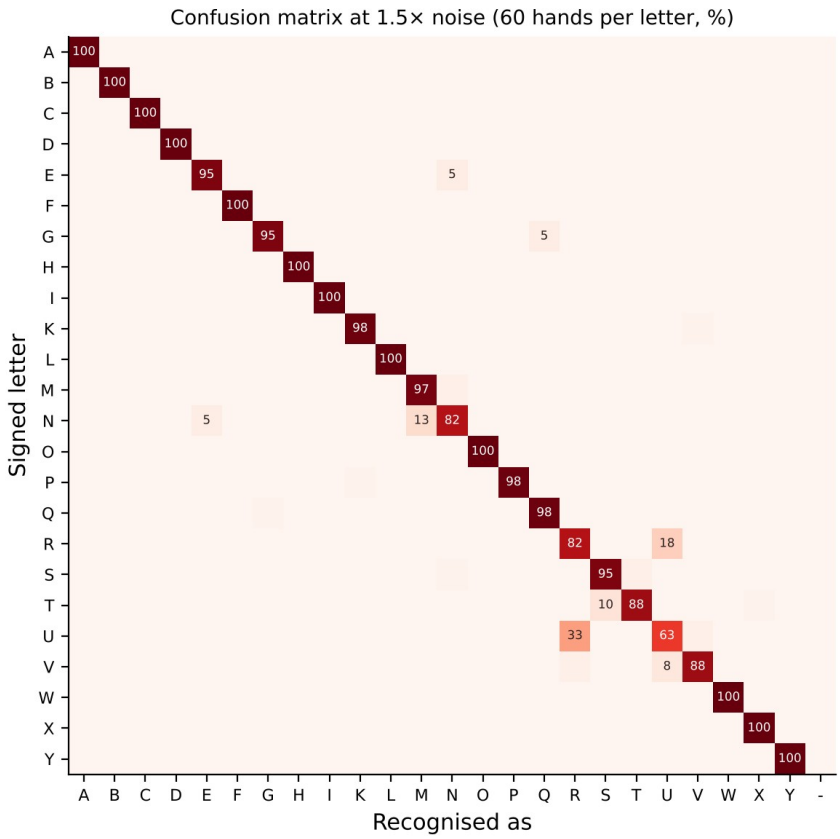


Figure 4. Confusion matrix of the static letters at 1.5× noise

4.3 Which Parts of the Design Matter (E2)

Table 8. Ablation of the feature design (1.5× noise, 15% wrong hand labels)

Design	Accuracy
Full design	96.0%
No mirror check (trust the hand label)	89.3%
No orientation features	83.5%
Camera-frame coordinates (no palm frame)	75.4%
One example per letter	95.4%

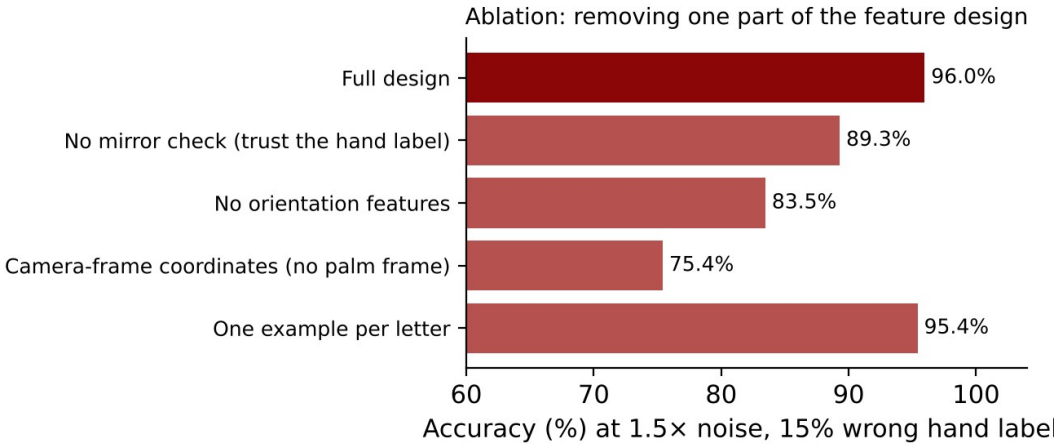


Figure 5. Accuracy when one part of the design is removed

The full design reached 96.0% under these hard conditions. Using camera coordinates instead of the palm frame dropped accuracy to 75.4%, which confirms Proposition 1 in practice: the palm frame removes most of the variation caused by how the hand is held. Removing the orientation terms gave 83.5%, because letters such as G and Q become identical, and trusting the camera's hand label instead of checking the mirror image gave 89.3%. Keeping only one template per letter still gave 95.4%, which shows that most of the accuracy comes from the geometry of the features rather than from the number of examples.

4.4 Moving and Two-Handed Signs (E3)

Table 9. Detection of moving and two-handed signs (30 trials each)

Sign	Type	Detected	95% Wilson CI
hello	moving	30/30	89% – 100%
yes	moving	30/30	89% – 100%
no	moving	30/30	89% – 100%
please	moving	30/30	89% – 100%
sorry	moving	30/30	89% – 100%
thankyou	moving	30/30	89% – 100%
where	moving	30/30	89% – 100%
water	moving	30/30	89% – 100%
eat	moving	30/30	89% – 100%
bathroom	moving	30/30	89% – 100%
J	moving	30/30	89% – 100%
Z	moving	30/30	89% – 100%
Ñ	moving	30/30	89% – 100%
love	two hands	30/30	89% – 100%
more	two hands	30/30	89% – 100%
friend	two hands	30/30	89% – 100%
name	two hands	30/30	89% – 100%
house	two hands	30/30	89% – 100%
help	two hands	30/30	89% – 100%

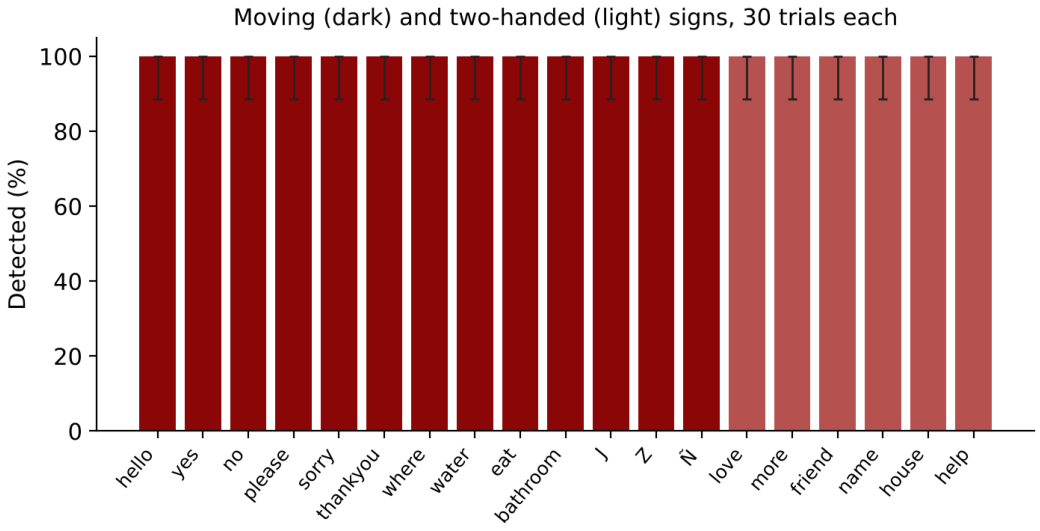


Figure 6. Detection rate of each moving and two-handed sign

All 570 trials were checked; 570 were detected (100.0%, 95% CI 99.3%–100.0%), and $H_0 2$ was rejected ($p < 0.001$). None of the 144 still-letter streams raised a moving sign (95% CI 0–2.6%), and the first letter written was correct in 100.0% of them. The experiments also found two faults that were then fixed: a waved hello was sometimes taken for the circle of please, because a straight back-and-forth path winds around its own centre, and the movement during a change from I to another letter was sometimes read as J. Adding the roundness condition of Equation 4 and requiring the handshape to be held throughout J, Z and Ñ removed both errors. This shows the value of testing decision rules against generated data before release.

4.5 Fingerspelled Words and the Hold Time (E4)

Table 10. Fingerspelled words read exactly (40 words per cell)

App hold time	Signer holds 500 ms	Signer holds 700 ms	Signer holds 1,000 ms
300 ms	88% (CER 2.4%)	95% (CER 0.9%)	92% (CER 1.4%)
400 ms	85% (CER 2.8%)	88% (CER 2.4%)	95% (CER 0.9%)
600 ms	0% (CER 46.2%)	80% (CER 3.8%)	88% (CER 2.4%)

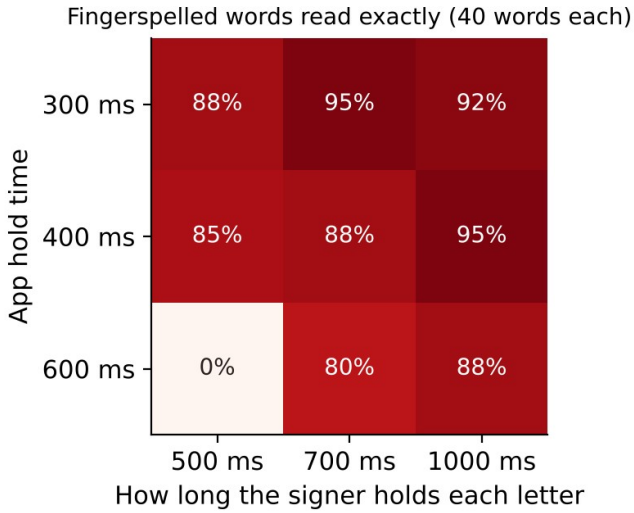


Figure 7. Word accuracy for each hold time and signer speed

At the default hold time of 400 ms, 107 of 120 words (89.2%) were read exactly across the three signer speeds, so $H_0 3$ (at most 80%) was rejected ($p = 0.006$). The table shows a clear trade-off. A hold time longer than the signer's own hold, such as 600 ms for a signer who holds letters for 500 ms, misses most letters; a short hold time suits fast signers but lets more transition shapes through. The default of 400 ms works for learners and fluent signers alike, and the app offers 300 and 600 ms for fast and slow signers. An earlier version used 550 ms and repeated a held letter on a timer; in this experiment it read only about half of the words, which led to the shorter default and to signing double letters with a slide, as fluent signers do.

4.6 Signs Taught by the User (E5)

Table 11. Recognition of signs taught from three recordings

Taught movement	Detected	Recorded as moving
spiral	30/30	3 of 3
climb	30/30	3 of 3
swipe	30/30	3 of 3
knock	30/30	3 of 3
Look-alike trials (built-in signs)	0 false accepts / 104	—

After three recordings each, the four new movements were found in 120 of 120 new trials, and none of the 104 built-in signs was mistaken for a taught one. A fixed dynamic-time-warping threshold had first missed the spiral in every trial and the swipe in most, because these movements naturally vary more in size; scaling the threshold by how much the user's own three

takes differ from each other (Section 3.4.6) solved this without adding false accepts. This lets Deaf users add local FSL signs and name signs that no dataset contains.

4.7 Braille Reader (E6)

Table 12. Braille reading accuracy by photo condition

Condition	Photos	Read exactly	CER
All photos	300	81.7%	12.6%
Printed dots	141	80.1%	15.3%
Embossed dots	159	83.0%	10.2%
Skew 0–2°	96	89.6%	6.3%
Skew 2–4°	100	77.0%	16.2%
Skew 4–6°	104	78.8%	14.6%
Blur < 0.8 px	149	80.5%	13.1%
Blur ≥ 0.8 px	151	82.8%	12.1%
Dot pitch < 15 px	107	77.6%	16.7%
Dot pitch 15–18 px	119	84.0%	11.8%
Dot pitch ≥ 18 px	74	83.8%	8.0%

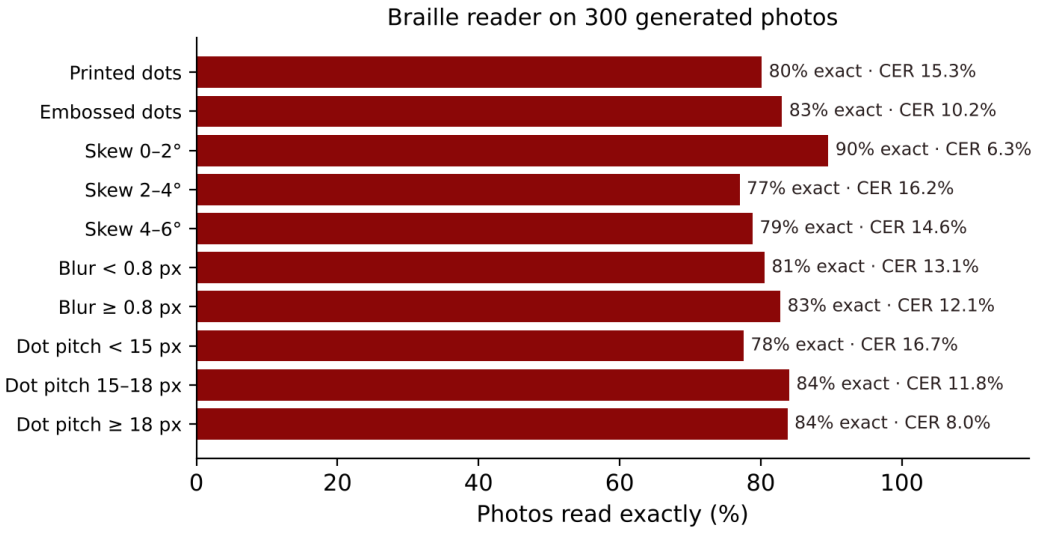


Figure 8. Share of braille photos read exactly, by condition

Across 300 photos the reader returned the exact text for 81.7% with a character error rate of 12.6%; $H_0 4$ was rejected. Embossed dots (83.0% exact) were read at least as well as printed dots (80.1%), because the reader chooses the polarity that forms a better grid. Larger dots helped (83.8% exact at a spacing of 18 pixels or more against 77.6% below 15 pixels), so users are told to hold the phone close. Most remaining errors came from lines whose first cell had dots only in its right-hand column, which can shift the fitted grid by half a cell.

Table 13. Ablation of two optional steps in the braille reader

Version	Read exactly	CER
Shipped reader	81.7%	12.6%
+ box smoothing	80.3%	14.8%
+ neighbour filter	81.0%	12.7%
+ both steps	79.3%	14.9%

The first version of the reader read 57% of the same photos exactly. Three changes raised this to the present level: a minimum blob size, a dot size estimated from the larger blobs, and choosing the polarity by neighbour support. Two further steps, extra smoothing and removing blobs without a neighbour, were also tested (Table 13). They did not improve accuracy, so they were left out of the app, which also makes it faster. Reporting this negative result keeps the design as simple as the evidence allows.

4.8 The Full Camera Path (E7)

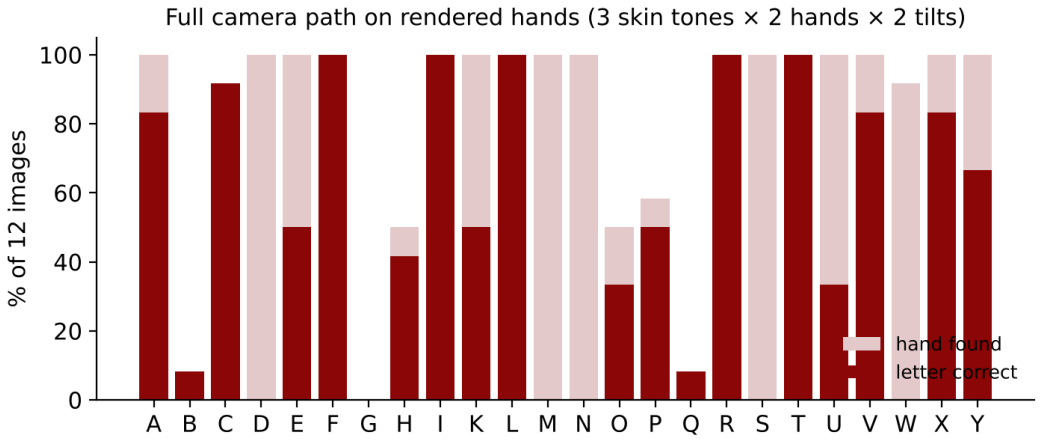


Figure 9. Rendered hands found by MediaPipe and letters named correctly

MediaPipe found a hand in 235 of 288 rendered images (81.6%), and the recogniser named 142 of those correctly (60.4%). The renders are flat drawings without skin texture, creases or nails, so the landmark model is working outside the kind of image it was trained on. This experiment is therefore a check that all parts fit together and a lower bound on the full path, not a measure of accuracy on real hands. The gap between this figure and E1 identifies the next step: calibrating the templates with real landmark recordings, which the app already supports through personal examples.

4.9 Speed (E8)

Table 14. Median processing time per frame or photo

Component	Median time	Frames per second it allows
MediaPipe hand landmarks (CPU)	71.0 ms	14
SENYAS recogniser and decisions	0.09 ms	11,161
Braille reader (one photo or frame)	25.6 ms	39

The recogniser added less than a tenth of a millisecond to each frame. Hand tracking dominated at 71 ms on a CPU, under the 100 ms limit, so $H_0$5 was rejected; on phones the model can use the graphics processor and usually runs faster. The braille reader scans a live frame about every 0.65 s, well within its cost.

4.10 Interface Fit, Accessibility and Coverage (E9)

Table 15. Layout test on 11 screen sizes (five screens each)

Screen	Size (px)	Layout	Sideways scroll	Small targets
Small phone	320×568	bottom tabs	none	0
Android phone	360×740	bottom tabs	none	0
iPhone	390×844	bottom tabs	none	0
Large phone	412×915	bottom tabs	none	0
Phone, sideways	844×390	side rail	none	0
Small tablet	768×1024	side rail	none	0
Tablet	820×1180	side rail	none	0
Tablet, sideways	1180×820	side rail	none	0
Laptop	1366×768	side rail	none	0
Desktop	1920×1080	side rail	none	4
TV / 4K (scaled)	2560×1440	side rail	none	0

On all 11 screen sizes, none of the five screens scrolled sideways and the five-tab navigation was always visible, so $H_0$6 was rejected. Phones used a bottom tab bar; phones held sideways, tablets and computers used a side rail; wide screens showed both translation directions side by side. The first layout test found many touch targets below 32 pixels in the phrasebook and settings; after they were enlarged to at least 40 pixels, only a few small controls remained on one size, in line with WCAG guidance on target size (World Wide Web Consortium, 2018).

Table 16. Text contrast against WCAG 2.1 (minimum 4.5 : 1)

Colour pair	Contrast	WCAG AA
Light: body text on page	16.7 : 1	pass
Light: muted text on card	5.6 : 1	pass
Light: button text on blue	6.0 : 1	pass
Dark: body text on page	16.1 : 1	pass
Dark: muted text on card	7.1 : 1	pass
Dark: button text on blue	6.9 : 1	pass

Camera caption on glass	12.2 : 1	pass

Every colour pair in light and dark themes met the 4.5 : 1 contrast required for normal text. The lexicon holds 93 words and the numbers 0–10; all 32 phrasebook sentences produced 63 signs in each language with no word left to fingerspelling, so a Bikol speaker, a Filipino speaker and an English speaker receive the same signs.

Table 17. Evaluation mapped to ISO/IEC 25010 quality characteristics

Characteristic	Evidence in this study
Functional suitability	E1–E7: accuracy of signs, words, taught signs and braille
Performance efficiency	E8: time per frame; 190 KB app page
Compatibility	Runs in browsers on Android, iOS, Windows, macOS and Linux, and as an Android app
Usability	E9: layouts, touch targets, contrast; three languages; reduced motion and large text
Reliability	Works offline after the first start; still-hand false alarms 0% (E3)
Security	No accounts; video, photos and audio never leave the device; consent before storage
Maintainability	Small modules (hand, avatar, recogniser, braille, data, app) with benchmark scripts
Portability	One codebase for web, installable app and Android

4.11 Summary of Hypothesis Tests

Table 18. Summary of hypothesis tests

Hypothesis	Result	Decision
$H_0$1 accuracy ≤ 90%	99.0%, $p < 0.001$	Rejected
$H_0$2 detection ≤ 90%	100.0%, $p < 0.001$	Rejected
$H_0$3 word accuracy ≤ 80%	89.2%, $p = 0.006$	Rejected
$H_0$4 CER ≥ 20%	CER = 12.6%	Rejected
$H_0$5 time > 100 ms	71 ms + 0.09 ms	Rejected
$H_0$6 layout faults on some screen	11 of 11 sizes without faults	Rejected

4.12 Limitations

The accuracy figures come from generated hands, movements and photos, which follow the shapes and noise levels described in Section 3.5 but cannot capture every real signer, camera and page. The rendered-hand test shows that real-image calibration is the largest remaining step. The sign vocabulary uses forms shared by FSL and ASL; regional FSL signs must be added by users or by future work with the Deaf community. Bikol is spoken with a Filipino voice because no Bikol voice is available on phones. Braille reading is limited to Grade 1.

CHAPTER 5. CONCLUSIONS AND RECOMMENDATIONS

5.1 Conclusions

1. A recogniser that generates its examples from a robot-like kinematic hand model and compares palm-frame features classified the static FSL letters with 99.0% accuracy at expected noise, without any recorded dataset. The palm frame, orientation terms and mirror check each contributed measurably.
2. Hysteresis counting, winding numbers, held-shape paths and two-hand distances detected 100.0% of moving and two-handed signs with no false alarms on still hands.
3. With a 400 ms hold time and double letters signed with a slide, 89.2% of fingerspelled words were read exactly across slow, average and fast signers.
4. Signs taught from three recordings were recognised in 100.0% of trials with no false accepts, so users can extend SENYAS with local signs.
5. Grid fitting read 81.7% of braille photos exactly (12.6% CER) across print, embossing, skew and blur.
6. Every component ran fast enough for live use on the device and offline, and the interface fitted every screen tested with readable contrast.
7. A single trilingual index carried every phrase into signs in English, Filipino and Bikol, and the Read aloud switch lets each user decide whether recognised words are spoken.

Overall, the study shows that an intelligent machine built on kinematics and geometry can provide an offline, trilingual communication aid for Deaf, blind and hearing people without big data, and that its behaviour can be measured and improved.

5.2 Recommendations

- Validate SENYAS with Deaf and blind learners, teachers and FSL interpreters in Albay, for example in SPED centres and with the Philippine Federation of the Deaf, under ethics review, using the Technology Acceptance Model (Davis, 1989) and task success rates.
- Record real landmarks from many signers to calibrate the templates, and compare the geometric recogniser with a small learned model trained on those recordings.

- Ask native Bikol speakers and Deaf FSL users to review every word and sign, and add regional FSL variants and Bikol phrases for local services and disasters.
- Add facial landmarks for FSL grammar, sentence-level translation, and Grade 2 braille.
- Develop a Bikol text-to-speech voice, and distribute SENYAS through the Department of Education, local disaster risk reduction offices and health centres.
- Repeat the speed and battery tests on low-cost Android phones.

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APPENDICES

Appendix A. Braille cells decoded by SENYAS

Table A1. Grade 1 braille cells read by SENYAS (dot 1–3 left column, 4–6 right)

Letter	Dots	Cell	Letter	Dots	Cell	Letter	Dots	Cell
A	1	⠁	K	1-3	⠅	T	2-3-4-5	⠞
B	1-2	⠃	L	1-2-3	⠇	U	1-3-6	⠥
C	1-4	⠉	M	1-3-4	⠍	V	1-2-3-6	⠧
D	1-4-5	⠙	N	1-3-4-5	⠝	W	2-4-5-6	⠺
E	1-5	⠑	Ñ	1-2-4-5-6	⠻	X	1-3-4-6	⠭
F	1-2-4	⠋	O	1-3-5	⠕	Y	1-3-4-5-6	⠽
G	1-2-4-5	⠛	P	1-2-3-4	⠏	Z	1-3-5-6	⠵
H	1-2-5	⠓	Q	1-2-3-4-5	⠟	Number sign	3-4-5-6	⠼
I	2-4	⠊	R	1-2-3-5	⠗	Capital sign	6	⠠
J	2-4-5	⠚	S	2-3-4	⠎			

Appendix B. Signs the camera reads directly

Letters A–Z and Ñ (J, Z and Ñ by their movement); numbers 0–10; words: hello, yes, no, thank you, please, sorry, where, water, eat, bathroom, I love you, good (thumbs-up), bad (thumbs-down), okay, love, help, more, house, friend, name, finished. Any other sign can be taught by the user from three recordings.

Appendix C. Trilingual lexicon

Table C1. Signed words in the three languages

English	Filipino	Bikol	Group
hello	kumusta	kumusta	greet
goodbye	paalam	paaram	greet
thank you	salamat	Dios mabalos	greet
please	pakiusap	tabi	greet
sorry	pasensya na	pasensya na	greet
yes	oo	iyo	greet
no	hindi	dai	greet
you're welcome	walang anuman	daing anoman	greet
fine	mabuti	marhay man	greet
nice	maganda	magayon	greet
meet	makilala	mamidbid	greet
excuse me	paumanhin	makiagi tabi	greet
I / me	ako	ako	people
you	ikaw	ika	people
we	tayo	kita	people
my	akin	sako	people
mother	nanay	nanay	people
father	tatay	tatay	people
brother / sister	kapatid	tugang	people
baby	sanggol	aki	people
friend	kaibigan	katood	people

family	pamilya	pamilya	people
name	pangalan	pangaran	people
Deaf	bingi	bungog	people
hearing	nakaririnig	nakakadangog	people
teacher	guro	paratukdo	people
learn	mag-aral	mag-adal	people
doctor	doktor	doktor	people
police	pulis	pulis	people
eat	kain	kakan	needs
food	pagkain	kakanon	needs
drink	inom	inom	needs
water	tubig	tubig	needs
bathroom (CR)	CR / banyo	CR / banyo	needs
sleep	tulog	turog	needs
home	bahay	harong	needs
house	bahay	harong	needs
school	paaralan	eskwelahan	needs
help	tulong	tabang	needs
more	dagdag	dagdag	needs
finished	tapos na	tapos na	needs
want	gusto	gusto	needs
need	kailangan	kaipuhan	needs
money	pera	kuwarta	needs
work	trabaho	trabaho	needs
play	laro	kawat	needs
go	punta	lakaw	needs
come	halika	dali	needs
stop	tigil	ontok	needs
wait	sandali	halat	needs
again	ulit	giraray	needs
slow	dahan-dahan	hinay-hinay	needs
write	sulat	surat	needs
call (phone)	tawag	apod	needs
see	kita	hiling	needs
understand	maintindihan	masabotan	needs
know	alam	aram	needs
not	hindi	dai	needs
sign language	senyas	senyas	needs
love	pagmamahal	pagkamoot	feel
I love you	Mahal kita	Namumutan taka	feel
happy	masaya	maogma	feel
sad	malungkot	mamundo	feel
angry	galit	maanggot	feel
tired	pagod	pagal	feel
sick	may sakit	may hilang	feel
hungry	gutom	gutom	feel
thirsty	uhaw	uhaw	feel
good	mabuti	marhay	feel
bad	masama	maraot	feel
okay	ayos	sige	feel
pain / hurt	masakit	makulog	feel
beautiful	maganda	magayon	feel
what	ano	ano	quest

where	saan	hain	quest
who	sino	siisay	quest
when	kailan	nuarin	quest
why	bakit	taano	quest
how	paano	paano	quest
emergency	emerhensiya	emerhensiya	emerg
hospital	ospital	ospital	emerg
fire	sunog	kalayo	emerg
earthquake	lindol	linog	emerg
flood	baha	baha	emerg
typhoon	bagyo	bagyo	emerg
danger	panganib	peligro	emerg
medicine	gamot	bulong	emerg
now / today	ngayon	ngonyan	time
tomorrow	bukas	sa aga	time
yesterday	kahapon	kasuhapon	time
morning	umaga	aga	time
night	gabi	banggi	time
zero	sero	sero	num

Appendix D. Phrasebook

Table D1. Phrasebook in three languages and the signs used (gloss)

English	Filipino	Bikol	Signs
Good morning	Magandang umaga	Marhay na aga	GOOD MORNING
Good evening	Magandang gabi	Marhay na banggi	GOOD NIGHT
Thank you	Salamat po	Dios mabalos	THANKYOU
You're welcome	Walang anuman	Daing anoman	WELCOME
How are you?	Kumusta ka?	Kumusta ka?	HOW YOU
I'm fine	Mabuti ako	Marhay man ako	ME FINE
What is your name?	Ano ang pangalan mo?	Ano an pangaran mo?	YOU NAME WHAT
My name is	Ang pangalan ko ay	An pangaran ko si	MY NAME
Nice to meet you	Ikinagagalak kitang makilala	Maogma akong mamidbid ka	NICE MEET YOU
I am Deaf	Bingi ako	Bungog ako	ME DEAF
Please sign slowly	Pakidahan-dahan ang pagsenyas	Hinay-hinay tabi an pagsenyas	PLEASE SIGN SLOW
I don't understand	Hindi ko maintindihan	Dai ko nasasabotan	ME UNDERSTAND NOT
Please repeat	Pakiulit po	Giraray tabi	AGAIN PLEASE
Please write it down	Pakisulat po	Isurat mo tabi	WRITE PLEASE
Where is the bathroom?	Nasaan ang CR?	Hain an CR?	BATHROOM WHERE
I need help	Kailangan ko ng tulong	Kaipuhan ko nin tabang	ME NEED HELP
Call a doctor	Tumawag ng doktor	Mag-apod nin doktor	CALL DOCTOR
I am sick	May sakit ako	May hilang ako	ME SICK
Where is the hospital?	Nasaan ang ospital?	Hain an ospital?	HOSPITAL WHERE
I love you	Mahal kita	Namumutan taka	ILOVEYOU
I'm hungry	Gutom na ako	Gutom na ako	ME HUNGRY
I'm thirsty	Nauuhaw ako	Uhaw ako	ME THIRSTY
Let's eat	Kain tayo	Magkakan kita	WE EAT
See you tomorrow	Kita tayo bukas	Magkita kita sa aga	SEE YOU TOMORROW

Goodbye	Paalam	Paaram	GOODBYE
I'm sorry	Pasensya na	Pasensya na	SORRY
Please wait	Sandali lang po	Halat muna tabi	WAIT PLEASE
Stop!	Tigil!	Ontok!	STOP
Danger! Leave now!	Delikado! Umalis na!	Peligro! Lakaw na!	DANGER GO NOW
There is a fire!	May sunog!	May kalayo!	FIRE
Earthquake!	Lindol!	Linog!	EARTHQUAKE
Are you okay?	Ayos ka lang?	Marhay ka lang?	YOU OKAY

Appendix E. Algorithms

Algorithm E1. Fitting the braille grid in one line

```
δ ← dot spacing; C ← column centres of the line (clustered with gap 0.45δ)
for P from 2.1δ to 2.9δ in steps of 0.05δ:
    for x₀ from C₁ to C₁ + P in steps of P/24:
        E ← Σ_c min(|u − kP|, |u − δ − k'P|), u = c − x₀
keep (P, x₀) with the least E
for each dot: cell k, column (0 or 1), row ← round((y − top)/δ); set bit
3·column + row
decode the six-bit numbers left to right with the number and capital states
```

Algorithm E2. Teaching a sign

```
record three takes of 1.5 s; for each take keep features, wrist path and two-
hand flag
if the hand moved less than 1.1 palm-lengths: store the mean features as
templates
else: resample each take to 16 frames; τ ← 1.6 × mean DTW distance between the
takes (0.75 ≤ τ ≤ 1.6)
while translating: every 5 frames compare the last 1.6 s with each take; accept
if DTW/τ < 1
```

Appendix F. Screens on other devices

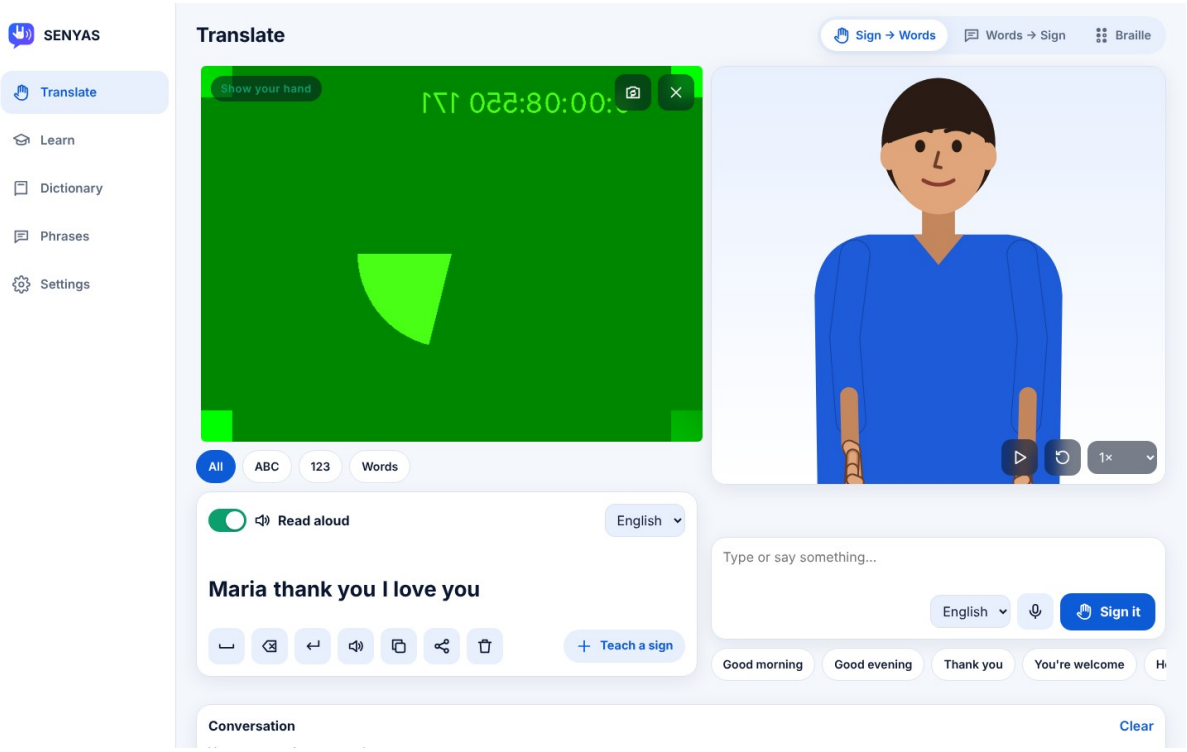


Figure F1. Desktop: both directions side by side as a conversation



Figure F2. Lessons, a lesson card in the three languages, and the dictionary

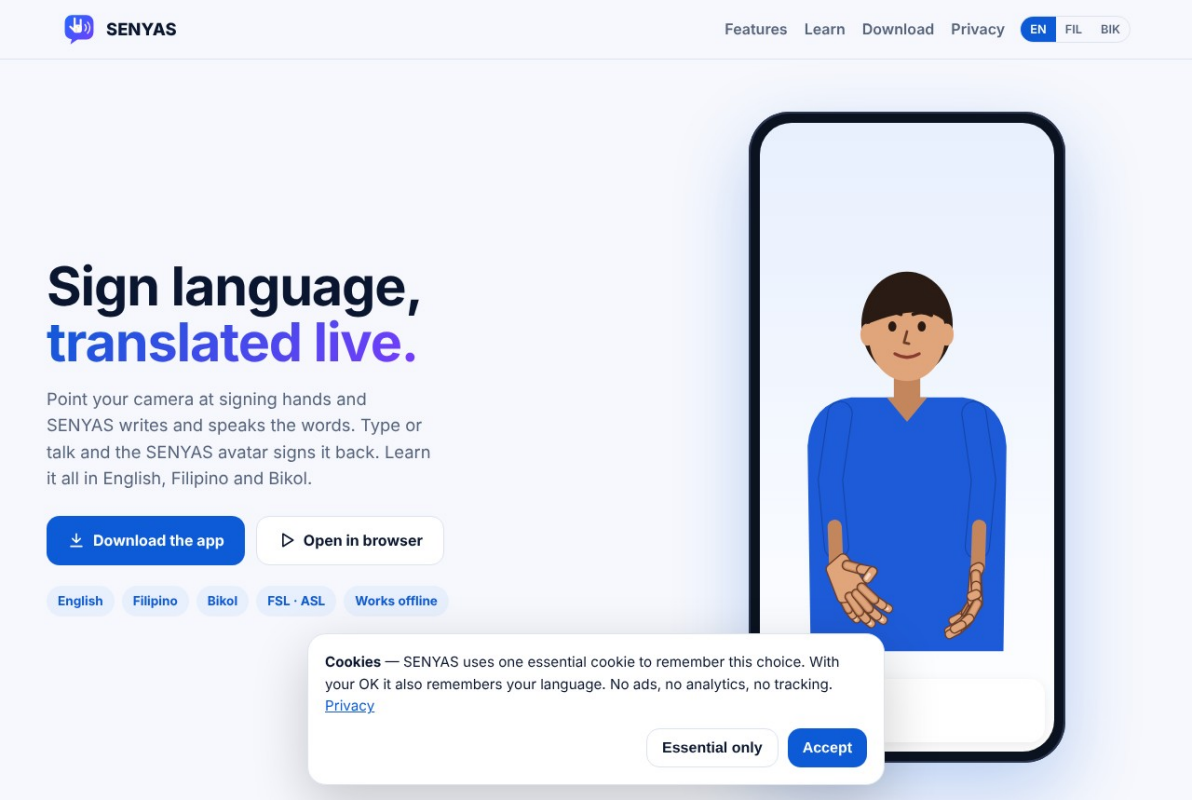


Figure F3. The SENYAS website with downloads and the cookie notice

Appendix G. User guide

1. Open SENYAS and choose what it may save on the device. Choose the app language and the language to translate into.
2. Translate → Sign → Words: press Start camera and sign in front of it. Hold each letter briefly; drop your hand to finish a word. Turn Read aloud on to hear each word, or off for silence.
3. Translate → Words → Sign: type or press the microphone and speak; the avatar signs it. Change the speed or tap a word to replay it.
4. Translate → Braille: upload a photo of braille or scan it live, holding the page flat and close; the text appears and can be spoken.
5. Learn: choose a unit, watch each sign, take the quiz, and practise with the camera. Teach a sign: record it three times.

Appendix H. Source code and data

The app, website, Android project, benchmarks and the scripts that produced every number, table and figure in this paper are in the SENYAS repository (github.com/pogi6969/SENYAS): src/ (hand.js, avatar.js, recog.js, braille.js, data.js, app.js), research/code/ (bench_recognizer.js, bench_browser.js, bench_page.html, bench_ui.js, analyze.py, and their results) and research/paper/ (this paper's generator).